

Chennai 2015: A novel approach to measuring the impact of a natural disaster

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Chennai floods

- Chennai is the capital of the southern state, Tamil Nadu
- One of the four important “metro” cities in India
- In 2015, it received a heavy spell of rainfall resulting in floods

9-10 Nov	First spells of rains in regions of Cuddalore
13 Nov	Floods in Kanchipuram
13-17 Nov	Floods in the low-lying areas of Chennai
1-2 Dec	Heavy rains and floods in Chennai

- Officially declared as a disaster area on the evening of 2 December.

Research questions

- How does the flood affect income and expenditure?
- What is the variation in expenditure by different types of households?
- What mechanisms are used to finance expenditure?

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 - God might have other plans
- Be there at high frequency when the flood comes.

Consumer Pyramids Household Survey

- A pan-India panel household survey of about 170,000 household
- Survey captures data on
 - Household demographics
 - Income
 - Expenditure
 - Household amenities, assets and liabilities
- The data is captured three times a year (known as a Wave or Round)
- One wave is completed over four months.

The survey questions

- **Data-table: Household Income & Expenses**

- Household level
- Estimates at the Rupee value.
- Maximum recall period of 4 months

- **Data-table: Household assets, amenities and liabilities**

- Binary variable: Yes/No
- Answered at the time of the survey
- Saved; Borrowed; Purchased assets
- Has outstanding savings
- Owns assets

Data

- **Treated:** HH in Chennai district (604 HH)
- Dropped districts in TN that saw a lot of rainfall.
- **Control:** HH in districts in Tamil Nadu other than Chennai and the dropped districts (5390 HH)

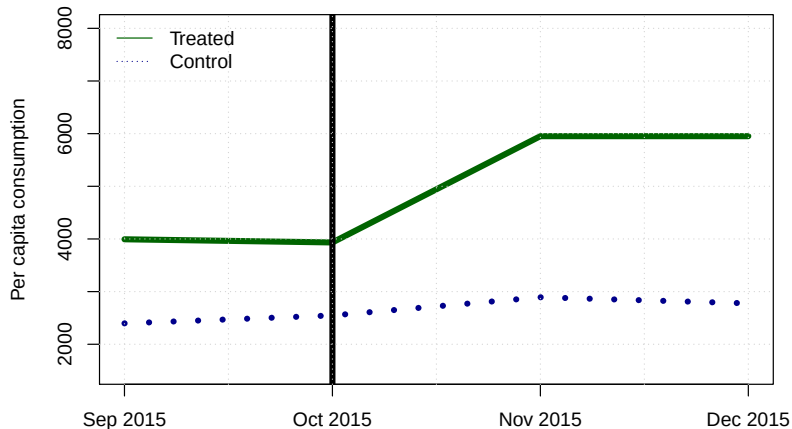
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- Focus on the *immediate* impact of the floods
- Pre-treatment: September and October, 2015
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- Fixed effects DID panel regression

$$y_{it} = \beta_0 + \beta_1 \text{treat}_{it} + \beta_2 \text{post}_{it} + \beta_3 \text{treat}^* \text{post}_{it} + \eta_i + \epsilon_{it}$$

Treatment effects: Per capita expenditure



Impact on income and expenditure

	log (per capita inc) (1)	log (per capita exp) (2)
TIME	0.049*** (0.006)	0.134*** (0.004)
TIME:TREAT	−0.008 (0.010)	0.280*** (0.014)
Observations	14,912	14,912
R-square	0.0049	0.23
<i>Note:</i>	Household FE	
<i>Note:</i>	Heteroscedasticity consistent SE	

Impact on components of expenditure

	log(Per capita expenditure)		
	Food	Health	Power/fuel
	(1)	(2)	(3)
TIME	0.058*** (0.002)	0.049*** (0.011)	0.037*** (0.013)
TIME:TREAT	0.030*** (0.006)	1.371*** (0.075)	0.520*** (0.034)
Observations	14,912	14,912	14,912
R-square	0.07	0.11	0.01
<i>Note:</i>	Household FE		
<i>Note:</i>	Heteroscedasticity consistent SE		

Heterogeneous treatment effects #1

	log(Per capita expenditure)		
	No concrete roof	No modern finance	Less than 50% mem. have mobile
	(1)	(2)	(3)
TIME	0.134*** (0.004)	0.134*** (0.004)	0.134*** (0.004)
TIME:TREAT	0.302*** (0.016)	0.338*** (0.018)	0.291*** (0.015)
TIME:TREAT:z	-0.121*** (0.026)	-0.142*** (0.025)	-0.059* (0.034)
Observations	14,912	14,912	14,912
R-square	0.23	0.24	0.23
<i>Note:</i>	Household FE		
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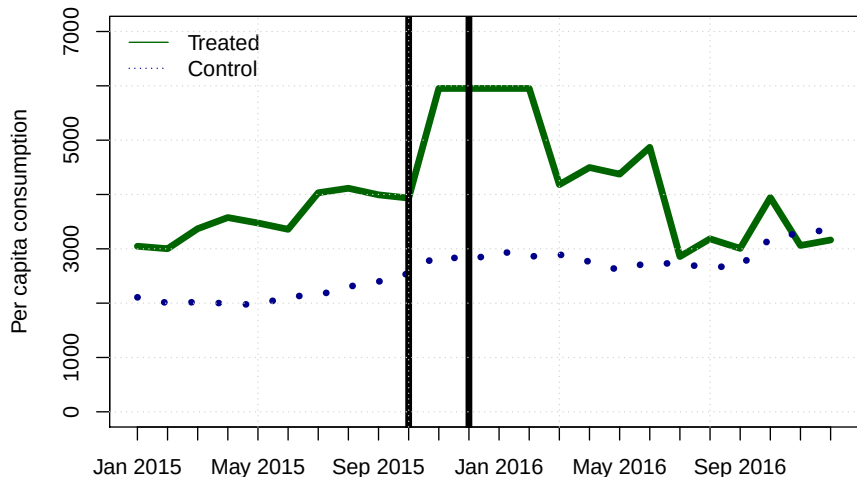
Heterogeneous treatment effects #2

	log(Per capita expenditure)		
	No AC	No vehicle	No fridge
	(1)	(2)	(3)
TIME	0.134*** (0.003)	0.134*** (0.003)	0.134*** (0.003)
TIME:TREAT	0.360*** (0.015)	0.299*** (0.011)	0.289*** (0.010)
TIME:TREAT:z	-0.139*** (0.027)	-0.190*** (0.030)	-0.145*** (0.040)
Observations	14,912	14,912	14,912
R-square	0.23	0.23	0.23
Note:	Household FE		
Note:	Heteroscedasticity consistent SE		

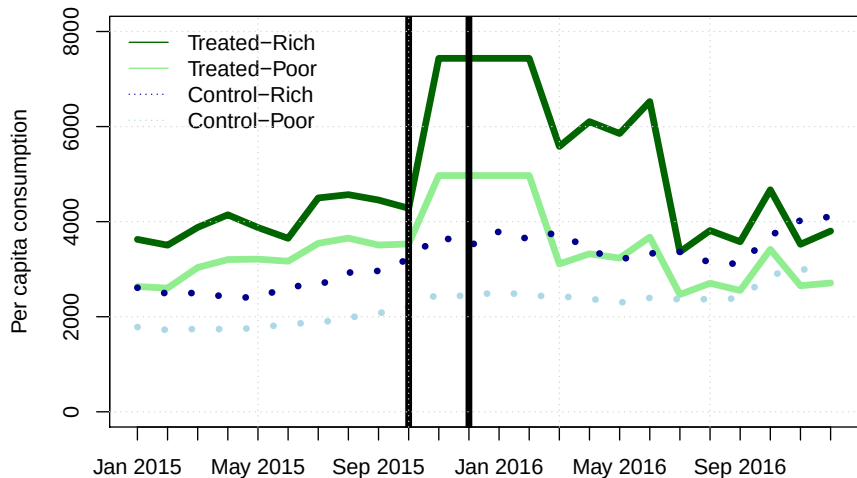
Impact on asset purchase, saving, borrowing

	Bought asset	Saved	Borrowed
TIME	0.060*** (0.005)	0.037*** (0.005)	0.176*** (0.006)
TIME:TREAT	-0.044*** (0.008)	-0.030*** (0.006)	-0.163*** (0.009)
Observations	14,912	14,912	14,912
R-square	0.02	0.01	0.11
Note:	Household FE		
Note:	Heteroscedasticity consistent SE		

Expenditure over a year



Expenditure over a year: By asset class



Robustness checks

- Matching so as to address the lack of match balance
- Placebo check (2014)
- Results go through

Conclusion

- The floods in Chennai led to a 32% increase in per capita expenditures
- The highest increase was on health, followed by power and fuel
- Vulnerable households also saw an increase in expenditure, but lower than that of the less vulnerable groups

Way forward

- We have a powerful new dataset for studying natural disasters
- This can be applied to all natural disasters in India
- The development of improved panel data can pave the way for a new level of understanding of natural disasters.
- Use of geo-location data to identify the extent of exposure of households to natural disasters

Questions/Comments